



Full-page recognition and alignment of historical musical documents

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Abstract

Optical Music Recognition aims to transcribe musical manuscript images into digital formats by using automatic methods for enhanced accessibility and preservation. This task is challenging for handwritten historical musical pieces from the Late Middle Ages, Early Renaissance, and previous time periods. This music has the interesting characteristic that both musical and lyrical elements are present with an implicit time alignment between them. This paper introduces techniques for simultaneously transcribing the musical and lyrical elements. We research how to automatically obtain the time alignment for an accurate musicological interpretation. Convolutional and Recurrent Neural Networks and Transformer models are explored for holistically transcribing and aligning historical pieces. This paper explores different techniques to improve the training of the models in limited data scenarios. Experiments are conducted on two different datasets from the same time period. Our findings highlight the potential of Transformer models in overcoming the alignment challenge, providing the best alignment capabilities without compromising the quality of transcriptions and offering a promising direction for future research in the automatic recognition of historical musical documents.

Keywords Optical Music Recognition · Handwritten Music Recognition · Transformer models · CRNN models · Aligned Music and Lyrics Transcription · Full Page Recognition

1 Introduction

There are millions of musical pieces from the Late Middle Ages, Early Renaissance, and earlier time periods that can only be found as original manuscripts. This vast number of pieces makes the task of transcribing them in substantial quantities with means other than automatic recognition unfeasible, given the time it would take for human experts to transcribe them. Historical musical pieces are particularly interesting because they do not originate in digital format. Optical Music Recognition (OMR) researches the automatic transcription of images of musical pieces to make them acces-

sible in a format other than simple images. Handwritten Music Recognition (HMR) is a more specific field of OMR that focuses on transcribing handwritten pieces. It emphasizes the need for adopting holistic approaches to deal with this problem instead of relying in a series of independent stages [1].

Many historical musical manuscripts have musical and lyrical elements in consecutive and alternate lines, as is shown in Fig. 1. Most of these historical musical manuscripts come from the religious tradition in which musical and lyrical elements were used in liturgical acts. For these musical pieces, these elements are explicitly synchronized at the line level with this synchronization being implicit in each line. Synchronization refers to a time alignment between musical and lyrical elements within each line. This synchronization is an essential part of the musical piece as it helps music experts to interpret the piece properly. It also represents the influence of music and lyrics on each other, which in itself is an interesting problem. Synchronization is very important for music information retrieval and for music and lyrics transcription. Fig. 1 shows an example of an aligned pair of music and lyrics.

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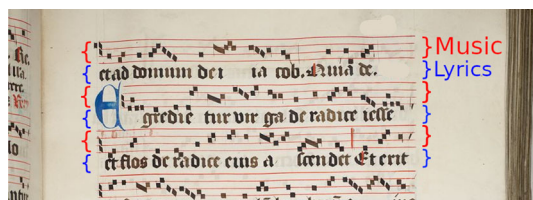


Fig. 1 Example of a historical musical piece from the Salzinnes dataset with alternating music and lyrics lines



Fig. 2 On the top, a music-lyrics pair from the Salzinnes dataset. On the bottom, that same pair with seven red bounding boxes representing alignments between musical notes and lyrical syllables

The synchronization between musical and lyrical elements is represented with syllables [2], where syllables correspond to musical patterns for musical elements and to linguistically motivated syllables for lyrical elements. An example of an aligned pair of music and lyrics is shown in Fig. 2. Alignment could be done with explicit alignment methods such as a naive one-to-one approach to alignment or using alignment models developed for machine translation. An alternate approach is for alignment to be done holistically by training models to perform both alignment and transcription of music and lyrics. This paper researches HMR with special interest in obtaining the time-synchronized alignment by recognizing the syllabic segmentation between lyrical and musical elements in a joint manner. This problem is known as Aligned Music Notation and Lyrics Transcription (AMNLT) [3]. The alignment problem has also been studied for the task of aligning the musical notation and an audio signal [4, 5]; however, our research focuses on the document images.

Taking into account the nature of the alignment between music and lyrics inside each line, a holistic approach seems to be the best course of action. Some papers have tried to solve the problem of alignment by combining the output of two recognition systems to produce an aligned result [6]. Convolutional and Recurrent Neural Networks (CRNN) are the state of the art in HMR [7]. They perform well even with small datasets; however, a real holistic approach to segmentation has been elusive until now.

Transformer models [8] are a promising approach for dealing with HMR given that they follow a segmentation-free approach. These models are based on attention mechanisms,

which point the model towards the important elements of images in order to enhance their recognition capabilities by acting as a mask that gives different levels of importance to different sections of documents. When using transformer models it is important to distinguish between global attention transformers and local attention transformers. Global attention transformers process whole sequences at once, while local attention transformers process a smaller attention window at a time.

Global attention transformers are better at capturing long-term relationships in sequences, but they are more costly regarding computation time and space. Local attention transformers focus on closer relationships between elements and are computationally more efficient. This makes local attention transformers more appropriate for this task since the language relationships that need to be modeled are within close proximity to one another. This is true even when working with long sequences (which is the case for HMR). These differences between attention types make local attention transformers more suitable for the task, given that we have limited training data.

As it happens with many machine-learning problems, one concern when working with automatic methods for HMR is the lack of data. Although there are millions of music pieces in archives, only a few are labeled adequately for OMR and HMR experiments and even fewer are labeled with alignment information. A usual solution to this problem is to perform some type of data augmentation. Although there are many ways to generate synthetic data, this data does not necessarily represent the real data distribution accurately, and even though digital images can be generated using historical data, they may not have the same optical characteristics as the original historical data. For this reason, working on models that are able to obtain good results with datasets of reduced size is of great importance.

Given that we work with scarce datasets, transformer-based systems may have a harder time adapting to this data since they need large quantities of data for their learning process. To overcome the data scarcity problem, we apply techniques such as hard negative mining and data augmentation techniques such as applying distortions to images.

The use of transformers and CRNN for HMR specifically for the AMNLT task is studied in this work, evaluating the capabilities of transcription and alignment of these models. This article is organized as follows: In Section 2 we review related research that forms the basis for this study. The technical framework is explained in Section 3. Our experiments and their results are shown in Section 4 and our conclusions are presented in Section 5.

2 Related work

OMR and HMR in the past have followed a series of consecutive and independent steps [9]: i) binarization of the images [10], ii) staff line separation [11], iii) note classification [12], and iv) the combination of the resulting elements [13]. This sequential approach can lead to accumulative errors due to the multiple steps involved, as errors produced in one step are transferred to the following steps.

Some of these processes were removed as the field matured since it was proven that they were not effective [14]. Later, holistic methods have proven to be more effective starting with the introduction of Hidden Markov Models [15] reducing the number of processing steps needed for the recognition of images.

More recent research has demonstrated that the use of CRNN-based systems [1, 16] is a more effective technique for OMR as a whole, and more specifically for HMR. These CRNN systems aim to perform as closely as possible as end-to-end processing of musical images, only requiring layout detection and image extraction as additional steps. Complex CRNN models such as the use of dense layers [7] and sequence-to-sequence models [17] have recently achieved better results in HMR tasks. CRNN-based systems are currently the state of the art in music recognition, as shown by the many successful works that use CRNN as a backbone, exploring music notation representation [18] and music structure representation [19].

Research in the field of HMR is mainly centered around the recognition of musical elements in documents. Only a small amount of research perform recognition of both music and lyrics simultaneously with regard to AMNLT. There are works that approach both music and lyrics, but they generally carry out this process in two separate tasks [20, 21]. The aim of the first works to leverage the combination of music and lyrics was to enhance the performance of the HMR systems and not to produce an alignment between the two parts [6]. Martinez-Sevilla et al. [3] were the first to approach the task in a completely coupled manner, producing not only the transcription of music and lyrics but also their alignment, doing so in a holistic manner. Some developments outside of the academic world with tools such as the Cantus Analysis Tool [22] and the OMMR4all tool [23] have shown an interest in approaching this task. However, their rule-based approach is not necessarily a generalizable tool.

Recently, encoder-decoder or transformer-based systems have been introduced for OMR, with results closing in on the state of the art and even improving it when considering the recognition of musical elements in both monophonic and polyphonic musical pieces. These approaches use reshaping methods to unfold musical images for recognition [24], take advantage of music notation and novel loss functions [25],

and face the task in a completely end-to-end manner going from a single image to a complete transcription [26].

The use of transformer-based systems for HMR is closely related to their use in Handwritten Text Recognition (HTR). Transformer architectures for HTR have been developed in various configurations, such as systems combining Convolutional Neural Network (CNN) encoders with transformer decoders and systems being composed of both transformer encoders and decoders.

A significant breakthrough was achieved using pre-trained models like TrOCR [27], which is a vision encoder-decoder transformer system that leverages the Vision Transformer (ViT) system [28] and Donut [29], a transformer vision encoder-decoder system that is trained end-to-end for text recognition tasks.

More specifically, Donut uses the Swin [30] architecture as its encoder, an enhanced version of the ViT. Swin's improved shifted window attention mechanism and hierarchical design enables it to handle large images effectively, while performing local attention. Its ability in processing complex visual information makes it very suitable for the AMNLT challenge where it is important to take into account two intertwined streams of information. Its end-to-end training approach simplifies the workflow, eliminating the need for intermediate segmentation steps.

3 OMR framework

This section introduces the main characteristics of the systems that were used in this paper for both line-level and page-level recognition and alignment. We explain how the training and inference processes were carried out and the techniques that were applied during them.

3.1 CRNN-based system description

The CRNN architecture used in this paper combines convolutional layers for spatial feature extraction with recurrent layers for sequence modeling following the architecture presented in [31]. The convolutional part of the network captures local visual patterns from the input images, such as note shapes or syllabic contours. It is composed of four convolutional layers with 16, 24, 48, and 96 filters, respectively, each followed by a max pooling layer. These convolutional layers are followed by recurrent layers that model the temporal dependencies across the sequence of extracted features. These layers are composed of three Bidirectional Long Short Term Memory (LSTM) layers with a 50% dropout [32]. Additionally, the network has a 50% linear dropout in the output layer.

An alternative approach to the CRNN architecture that has been explored in this work involves replacing the simpler

CNN component previously described with a ResNet [33] backbone. The ResNet backbone offers the advantage of being a more expressive and pre-trained feature extractor, capable of capturing complex visual patterns. However, its increased complexity requires a more intensive fine-tuning process to adapt to the specific characteristics of musical notation. In this study, both configurations, the model with a basic CNN backbone and the model incorporating a ResNet backbone, were implemented and compared.

The CNN part of the network was trained with the Back-Propagation algorithm [34], whereas the RNN part was trained with the Back Propagation Through Time algorithm [35] using the CTC loss function [36]. This function allowed the model to learn mappings between input sequences and unsegmented target labels. The network was trained with synthetic images in addition to real historical images. These images were produced by taking Latin chants from the same period of the datasets and applying a syllabification process to obtain lyrics transcripts. An n-gram language model trained with music syllables from the training partitions was used to generate music transcripts. Then, images were generated based on these texts by using the `trdg`¹ tool for lyrics images and by concatenating musical image parts for music images following the procedure performed in [31]. Additionally, data augmentation such as distortions was applied during the training of the optical model (the visual recognition component of the system). The optical model was combined with an N-gram Language Model for the decoding phase.

The objective posed in this research is to obtain aligned transcriptions by performing the simultaneous recognition of both the musical and lyrical components in a musical piece. CRNN-based systems like the ones introduced above have been extensively used for sequence recognition by using the CTC loss function. However, they have rarely been researched to deal with structured problems where vertical relations are important, with the exception being in the field of Mathematical Expression Recognition [37].

This problem is not so relevant with transformer-based systems since their lack of explicit spatial restrictions in the inference process allows them to transcribe elements that are present simultaneously in a time frame.

3.2 Donut-based system description

Donut [29] is a vision encoder-decoder transformer model consisting of a pre-trained Swin model [30] as the encoder and the pre-trained BART text transformer component [38] as the decoder. Donut was trained in two stages to deal with document understanding tasks. In the first stage, the model was pre-trained to learn to read following the reading order by

using 11 million images of scanned documents in English. In the second stage, once the model learned to read, it was fine-tuned for musical document understanding. To leverage the general knowledge of the Donut model, we started from the weights obtained in the aforementioned first training stage.

Given that the base model only knows how to read scanned texts in English, we had to adapt it to deal with historical manuscripts and the Latin language. This was done through a fine-tuning process, for which we used grid search to determine the optimal learning rate, optimizer, and number of epochs in order to achieving the most efficient and effective parameters. We found that a learning rate of $1e^{-5}$ with the AdamW optimizer [39] over 180 epochs allowed us to adapt the model to HMR, using the cross-entropy loss function.

This fine-tuning process was done at the syllable level, using the data from the training partitions of our datasets. The use of synthetic images has also been considered and tested for this process. We carried out experiments that used data from our datasets and digitally rendered images of data from Gregorian chants of similar nature in order to fine-tune the model. These digital images did not provide any improvements in the recognition results.

3.2.1 Additional training details

Different techniques were used to enhance the inference capabilities of the model. Teacher forcing is a method that is used in training sequence prediction models. Instead of using the prediction of the model as input for the next step, the ground-truth (GT) from the training data is used. This approach helps the model to learn more quickly by consistently providing the correct sequences during training. Although not always explicitly named, teacher forcing is commonly used in most transformer approaches for training (e.g. in [8]). Additionally, Hard Negative Mining (HNM), which involves repeating samples that produced the highest loss during each epoch taking into account the mean and standard deviation of the errors, along with gradient accumulation, tends to lead to better results. We applied HNM after each epoch and used a batch size of 1 with a batch accumulation of 4. Furthermore, full-page synthetic images were constructed, parting from the synthetic images generated for training the CRNN models. These images were arranged on an empty background matching the original color distribution to emulate the layout of a complete page. This aims to provide more spatial context during training, potentially improving the model's ability to learn the page-level structures.

Donut pre-training was performed using images with dimensions of $2560h \times 1920w$ pixels. Consequently, images were rescaled to these dimensions in the training, validation, and test partitions of the datasets. The maximum sequence length for each of the datasets is established as the length

¹ <https://github.com/Belval/TextRecognitionDataGenerator>

of the longest sequence found in the training and validation partitions in each of them.

An adjustment was necessary regarding the dimension of images for one of the datasets due to the limitations of the 24 GB graphics processing units used in this work, which required finding a balance between image resolution and the maximum sequence length. The fine-tuning of the Donut model was done over the training partitions of the different datasets that are introduced in Table 1.

3.2.2 Inference

During inference, Donut processes the input image through its encoder to capture the image visual features. The decoder then generates text auto-regressively, starting with a special start token. The model predicts the next token using a greedy decoding strategy by selecting the one with the highest probability at each step. This chosen token is added to the sequence, and the process is repeated. This continues until an end token is generated or a maximum sequence length is reached [29].

3.3 Differences between the CRNN and the donut systems

As mentioned in previous sections, line recognition systems that are usually based on CRNN models are compared with full-page recognition systems in this article. This means that line detection, segmentation, and extraction have to both be carried out for the CRNN models for training and inference. These detection, segmentation, and extraction processes were performed automatically using the detec-tron2 model presented by Wu et al. [40], which was trained using the training partitions of the corresponding datasets. This ensures a fair comparison with the segmentation-free approach from the Donut system.

Training at the line level has some advantages. First, the relationships that the model needs to learn are simpler. Second, there are more examples of the relationships that can be observed in these sequences. Lastly, the model only needs to learn to process images from left-to-right, and it does not explicitly process the vertical dimension other than by performing feature extraction. The main disadvantage is that there is information that is lost without the whole context of a document.

Training with full pages as input, which was performed with transformer-based systems in this work, also has its advantages. First, the amount of information that the system can use when analyzing a part of the image is greater. And second, it is not necessary to perform a layout analysis process, which helps to avoid the mistakes that can derive from this step. However, it uses notably more resources and faces problems such as having to learn the reading order of documents.

Table 1 Characteristics of the Einsiedeln and Salzinnes datasets

Datasets	Einsiedeln	Salzinnes
Number of pages	362	181
Number of pairs staff/lyrics	6125	2251
Number of music symbols	23	20
Number of lyrics symbols	52	46

In line-level processing, the reading-order problem is solved geometrically, once lines are segmented, they are read top-to-bottom and left-to-right. Some music images have narrative texts mixed in with the music and lyrics elements, which could present a challenge. However, these texts are to be read either before the elements of the musical piece they accompany (if they appear at the left margin) or after (if they appear at the right margin).

The reading-order problem is mainly relevant when full-page recognition is performed. The transformer-based system is expected to learn in what order elements should be read, whether these are musical, lyrical, or narrative elements. For the model to be able to learn how to process the documents in the expected reading order the GT must include this reading-order information. It is important to remark that it was not necessary to apply curriculum learning [41] in order for the model to learn the reading order. Presenting the data in the correct reading order when training the model was enough information for it to be able to solve this problem.

4 Experiments

4.1 Corpora

Two different corpora were used in this study, the Salzinnes [42] and the Einsiedeln [43] datasets, dating from the 15th century. Fig. 3 shows examples and Table 1 shows the main characteristics of these datasets. These datasets included both images and GT.

In datasets of this period, notes were only distinguished by their pitch and clef and not by their shape, with their duration being implicit in the lyrics. Thus, with this notation, the number of symbols with different semantic meaning is the number of possible positions of the scores times the number of possible positions of the clefs times the number of different clefs. This can be a problem when the amount of training data is limited.

In this study, as we processed the transcriptions, we changed from using musical notation GT to geometrical notation. In geometrical notation, each music symbol is transcribed as the position it occupies on the staff. This step removes the visual ambiguity by avoiding the situations where a note can be represented in different positions of

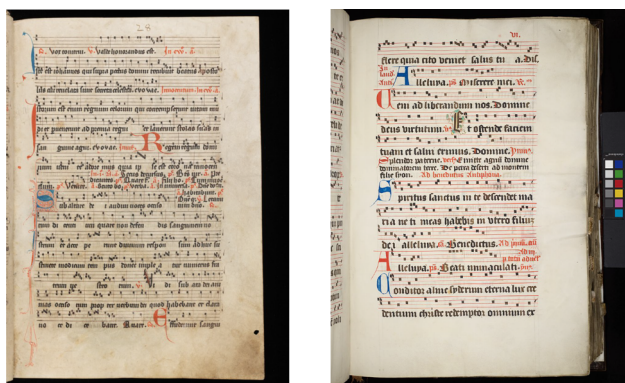


Fig. 3 A page from the Einsiedeln dataset is shown on the left, and a page from the Salzinnes dataset is shown on the right

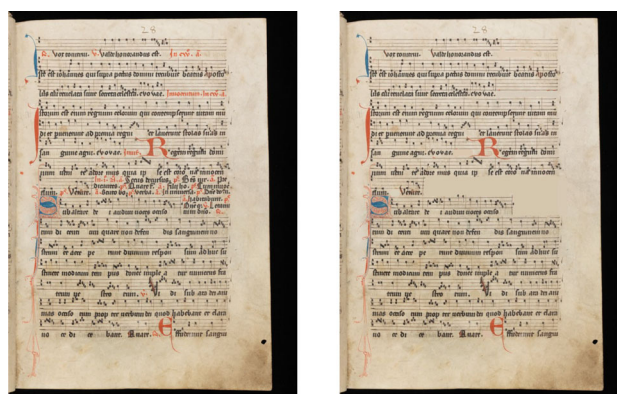


Fig. 4 Example image before and after removing the narrative elements for the Einsiedeln dataset

the staff depending on the clef. It increases the number of training data per symbol, but it also increases the semantic ambiguity.

The clef together with the geometrical position of symbols on the scores define a note, which is computed in this work as a posterior process. This allows us to obtain musical notes from geometrical positions. This method is entirely dependent on the clefs being properly recognized since they are necessary to be able to recover the musical note information. In a production environment, errors could be corrected by changing the individual clefs and then rerunning the post-processing.

Note that these experiments produced transcripts with geometrical information since the models were trained on geometrical positions on the image scores. However, for evaluation, we needed to deal with musical notes. To convert score positions into notes, we used the clef associated with each staff in the GT and applied a direct transformation. For example, the geometrical sequences "C4 S2 S2 L3 S4" and "F3 S3 L3 S3 L3" become "C4 g2 g2 a2 c3" and "F3 g3 f3 g3 f3" through a simple post-processing translation step.

Additionally to the geometrical and musical notations, in this paper we also explored the performance of the CRNN and Donut models with a third option regarding notation. This is the Humdrum Kern² musical notation. Kern is a symbolic music representation format that encodes musical information as plain text, where tokens separated by spaces or tabs represent specific musical attributes. This format is conceptually closer to traditional musical notation than to geometrical notation, yet its distinct approach to encoding musical elements makes it sufficiently different to warrant experimental evaluation.

With regard to the structure of the transcriptions at the line and page levels and at the character and syllable levels, we explored two different possibilities. First, we encoded music



Fig. 5 Example image before and after removing the narrative elements for the Salzinnes dataset

and lyrics notation separated in lines, where each music line in the transcriptions was processed before each lyrics line. Second, we encoded music and lyrics notation, with the information in the transcription being intertwined between music and lyrics. Both of these encodings follow the GT as established in the datasets, where divisions are present between musical notes and lyrical characters, and between syllables, allowing us to separate music and lyrics at both the line level and at the syllabic level.

Additional encoding structures could be considered, such as having a symbolic representation for music and lyrics separation within the syllables allowing the model to learn a more fine-grain representation. However, keeping the number of elements in the encoding to the necessary minimum appears to be the most effective approach since it simplifies the input and ensures that the model can learn without unnecessary complexity.

The datasets used in this article have narrative texts that are not annotated in the GT for training. These zones of the images were replaced with the background colour of their respective pages. This was done for images in the training, validation, and test sets of both of these datasets. Figures 4 and 5 show an example of this.

² <https://www.humdrum.org/rep/kern/>

Both datasets have been split into five partitions for cross validation, where 80% of the data was used for training and validation and the remaining 20% was used for testing. Recognition models were trained for each of the datasets independently.

The size of the training partitions may appear small, given the vast amount of resources usually needed to train Transformer models. Because of this, a digital dataset was used for pre-training to enhance performance. However since this approach did not lead to any improvements, the results obtained with this pretraining are not shown to avoid unnecessary numerical information.

4.2 Experimental setup

The GT of the datasets was used to perform experiments in two different ways. First, the musical and lyrical elements were recognized separately with two independent systems. Then, the alignment was derived either by pairing the different elements one-to-one in a naive approach or by using an explicitly trained alignment model. The IBM Model 2 [44] was used to obtain the alignment. This model is commonly used in monotonic sequence alignment tasks. To train the IBM Model 2, the pairs of lyrics-music alignments from the training partition were used in conjunction with the fast align method [45]. In the second approach, both staves and lyrics were jointly and synchronously recognized and an explicit alignment was obtained in the recognition process.

With regards to the recognition of the elements in the document, the following four experimental scenarios were defined (see Fig. 6 for examples):

- Separated lines scenario: Music and lyric parts are considered separately with two independent CRNN-based systems both for training and recognition. Line images and line transcripts are joined at the page level following the technique presented in [31]. In that technique, after segmenting and extracting the lines of a given page, these are joined together by pairing them into a single line following the reading order. This is done separately for music and for lyrics.
- Joined lines scenario: Music and lyric parts are jointly processed in a unique CRNN-based system for both training and recognition. Line images and line transcripts are joined at the page level, in the same manner as in the Separated lines scenario. Line transcripts (music and lyrics) are merged in just one line transcript. Note that the alignment is implicitly represented in the GT.
- Separated full-page scenario: A full image and the music and lyric transcripts are used in a transformer-based system for training. The transcripts of the music part and the lyric part are alternated in just one line transcript. The alignment is loosely represented in the GT.

- Joined full-page scenario: A full image and the music and lyric transcripts are used in a transformer-based system for training. Music and lyric transcripts are merged in just one line based on the GT alignment. The alignment is implicit in the line transcripts in this scenario.

The main reason for defining two different scenarios (one for each recognition model) was to study the differences between implicit and explicit alignments and to observe how reading order affects the recognition quality. The goal was to find out whether reading music and lyrics together for a holistic alignment performs better than reading them separately or alternately and aligning them afterwards.

The “<>” sequence in the transcripts shown in the GT in Fig. 6 was used to separate syllables. The first symbol of music transcriptions (C3 in this example) always refers to the musical clef.

In this paper, CRNN-based systems work at the line level, whereas transformer-based systems work at the full-page level. This may make a fair comparison between the two types of systems difficult. However, following the techniques presented in [31], where lines are joined to contain the information of a full-page, a fair comparison between the two models can be performed since they consider the same input.

The separated lines scenario is an easier problem for training since two separate CRNN-based systems are used. This is the most difficult scenario for correctly aligning music and lyric transcripts. The joined lines scenario stresses the unique CRNN-based system since the CRNN model has to take care of both transcripts; however the alignment is easier to obtain after the recognition process since it is implicit in the transcription. The separated full-page scenario forces the system to learn the reading order of the lines alternating between music lines and lyric lines. The alignment in this case is obtained after the recognition process with either of the posterior alignment methods. The joined full-page scenario forces the system to learn music and lyric lines simultaneously as a human being would perform. The alignment in this case is obtained easier than in the separated full-page scenario after the recognition process since it also is implicit in the transcription.

To be able to compare the methods that produce a holistic alignment in the recognition process from those that do not produce this alignment, we needed to define a way of obtaining this alignment. Note that the transcripts have the syllables segmented. The following two approaches were studied: first, the most straight-forward method to obtain the alignment was to pair syllables in the music and lyrics parts one to one using the “<>” symbol as an anchor and padding at the end in the case of a length mismatch. Second, in addition to this naive approach, an alignment model based on the IBM Model 2 was trained and used to produce an alignment.

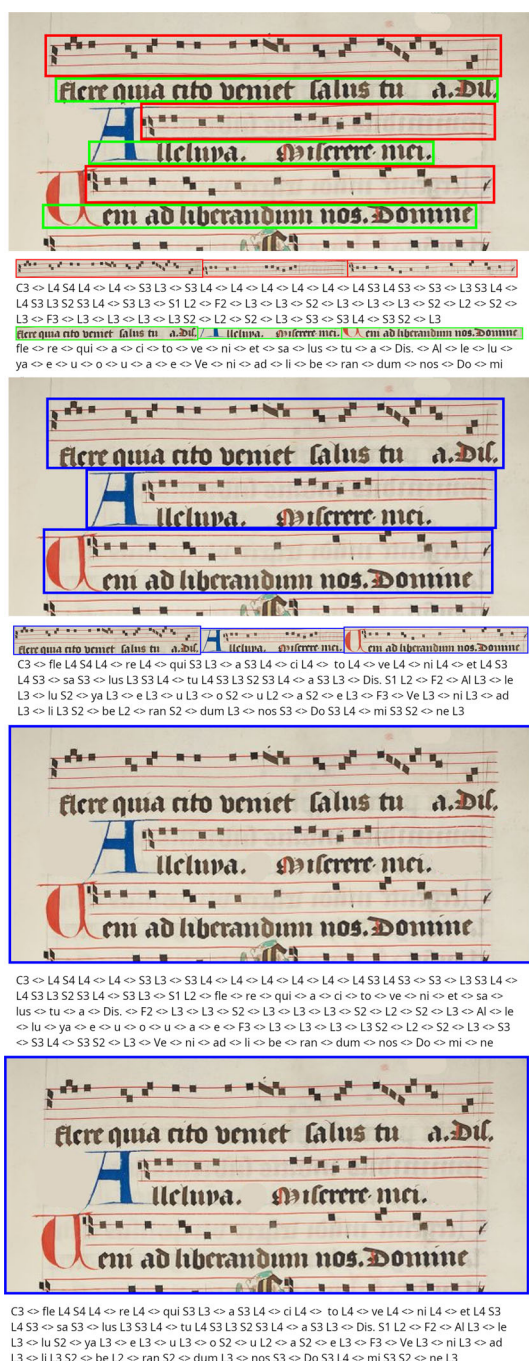


Fig. 6 Sample images and corresponding GT used in four scenarios. The first image shows the data used in the separated lines scenario. Red lines are joined as shown below the partial page image. The same for green lines. The second image shows the data used for the joined lines scenario. Blue lines are joined for the full page. The third image shows the data used in the separated full-page scenario. The last image shows the data used in the joined full-page scenario

Training and decoding were performed with free available tools³ for the separated lines scenario and the joined lines scenario. An N-gram language model was used where N was computed empirically over the validation sets.⁴ The optical models were trained as explained in [31] following the architecture shown in Section 3.1. The transcript results were then aligned using both of the methods presented in Section 4.2: the naive approach and the IBM model 2 approach. These experiments were considered as the baseline. A single experiment was performed in the joined lines scenario for each of the aligned datasets. The same CRNN architecture used in the separated lines scenario was considered in the joined lines scenario. For the separated full-page scenario, training and recognition were performed using the Donut model discussed in Section 3.2. The transcript results were aligned using both of the methods presented in section 4.2. The same Donut architecture from the separated full-page scenario was used for the joined full-page scenario.

4.3 Evaluation

Three metrics were used to evaluate the transcripts: Character Error Rate (CER), Diplomatic Symbol Error Rate (DSER) and Syllable Error Rate (SER).

CER calculates the percentage of characters that are incorrectly recognized by the model. DSER calculates the percentage of diplomatic symbols (in our case, musical symbols) that are incorrectly recognized by the model. SER calculates the percentage of syllables that are incorrectly recognized by the model, which is applicable for both characters and diplomatic symbols. While CER and DSER are conceptually similar, the standard practice in OMR research is to report CER for lyrics and DSER for musical notation separately. Although the underlying calculation is essentially the same, we adhere to this convention to maintain consistency with established methodologies in the field.

These metrics are calculated using the Levenshtein distance between the predicted text and the ground truth, at different levels of tokenization, divided by the total number of elements in the ground truth.

Given the nature of the transcriptions and alignments produced by the explicit segmentation methods (the joined lines and joined full-page scenarios) used in this work, it is not possible to separately evaluate the alignment error rate from the transcription error, as the alignment error is directly influenced by the quality of the transcriptions. Therefore, the alignment error rate was computed as the edit distance between the hypothesis and the GT at the syllable level. By

³ <https://github.com/jpuigcerver/PyLaia> <https://kaldi-asr.org/> <https://openfst.org/>

⁴ <http://www.speech.sri.com/projects/srilm/manpages/nggram-count.1.html>

Table 4 Alignment results obtained for each of the best performing experimental setups in the separated (S.) lines scenario, the joined (J.) lines scenario, the separated full-page scenario and the joined full-page scenario for the Einsiedeln dataset

Einsiedeln Scenarios & Alignment Methods	Align. Error
S. lines CRNN Synth+DA - Naive	26.4
S. lines CRNN Synth+DA - IBM M2	24.4
J. lines CRNN Synth+DA - Holistic	13.9
S. lines CRNN-RNET Synth+DA - Naive	27.3
S. lines CRNN-RNET Synth+DA - IBM M2	24.2
J. lines CRNN-RNET Synth+DA - Holistic	14.0
S. full-page Donut HNM+DA - Naive	27.6
S. full-page Donut HNM+DA - IBM M2	25.6
J. full-page Donut HNM+DA - Holistic	13.4
S. full-page Donut Synth HNM+DA - Naive	28.4
S. full-page Donut Synth HNM+DA - IBM M2	26.7
J. full-page Donut Synth HNM+DA - Holistic	14.1

working at the syllable level, with musical and lyrical syllables paired together, we measure how many complete units of alignment are correct when calculating this edit distance. This corresponds more closely to a total syllable error rather than a pure alignment error, however, it represents how many alignment units are correctly established by the model.

It must be noted that this is a pessimistic evaluation of the alignment, given that elements were only considered properly aligned if they were both accurately transcribed and aligned. Correct alignments with incorrect transcriptions were considered to be incorrect.

The evaluation process was carried out for the five partitions and a 95% confidence interval was calculated via bootstrapping. Finally, the average result of the five partitions was calculated.

4.4 Results

Tables 2 and 3 show the transcription results obtained in the experiments for all of the scenarios. Tables 4 and 5 show the alignment results obtained in the experiments for all of the scenarios.

The techniques presented in Section 3.2.1 were tested over the validation partitions of the datasets, showing that both HNM and data augmentation techniques helped improve the quality of the transcriptions produced, whereas the use of synthetic images did not have significant impact on the results. Tables 2 through 5 include results at the inference level both for when these techniques are considered and for when they are not applied. Reported results are compared to our previous results published in [31]. Note that other research with these datasets only provided experiments on layout analysis.

Table 5 Alignment results obtained for each of the best performing experimental setups in the separated (S.) lines scenario, the joined (J.) lines scenario, the separated full-page scenario and the joined full-page scenario for the Salzinnes dataset

Salzinnes Scenarios & Alignment Methods	Align. Error
S. lines CRNN Synth+DA - Naive	64.1
S. lines CRNN Synth+DA - IBM M2	56.9
J. lines CRNN Synth+DA - Holistic	19.9
S. lines CRNN-RNET Synth+DA - Naive	65.2
S. lines CRNN-RNET Synth+DA - IBM M2	58.3
J. lines CRNN-RNET Synth+DA - Holistic	20.1
S. full-page Donut HNM+DA - Naive	67.4
S. full-page Donut HNM+DA - IBM M2	59.9
J. full-page Donut HNM+DA - Holistic	14.1
S. full-page Donut Synth HNM+DA - Naive	69.7
S. full-page Donut Synth HNM+DA - IBM M2	62.1
J. full-page Donut Synth HNM+DA - Holistic	14.7

These experiments were also performed by directly using musical notation and by using humdrum kern notation instead of training and recognizing geometrical notation and performing post-processing to obtain musical notes before evaluation. Tables 6 and 7 show how the recognition of geometrical positions, which are subsequently transformed into musical notes, achieves significantly better results than the direct recognition of musical notes or kern notation. Results obtained using the ResNet backbone for the CRNN model and those incorporating synthetic images during Donut model training are not reported, as they lead to the same conclusions presented in the tables stated.

Finally, even though the narrative texts were removed for experimentation to make a fair comparison between page-level experiments and line-level experiments, as they are absent in line-level data and appear as noise in page-level images (not included in the ground truth), we show results obtained by training the Donut model on images containing narrative texts to assess their impact on recognition performance when these texts are regarded as noise in Tables 8 and 9.

The following aspects are noticeable in the results shown in this section:

- Data augmentation and hard negative mining techniques perform better when applied together. The most important of these techniques is data augmentation. It can be observed that not using either of the techniques is better than only using hard negative mining. The reason for this is that without data augmentation, hard negative mining causes over-fitting in the model over the training data.

Table 2 Lyrics and music transcription results obtained in the separated lines scenario, the joined lines scenario, the separated full-page scenario and the joined full-page scenario for the Einsiedeln dataset

Einsiedeln Scenarios	Lyrics Error Rate		Music Error Rate	
	CER	SER	DSER	SER
Separated lines CRNN Synth+DA	3.8 ±1.0	8.3 ±1.6	2.9 ±0.8	6.7 ±1.3
Separated lines CRNN-RNET Synth+DA	3.9 ±0.9	8.5 ±1.6	3.1 ±0.9	6.5 ±1.3
Separated lines Donut Synth+HNM+DA	5.4 ±1.1	10.2 ±2.3	4.2 ±1.1	7.7 ±1.9
Joined lines CRNN Synth+DA	5.7 ±1.4	10.8 ±2.4	3.9 ±1.0	6.3 ±1.6
Joined lines CRNN-RNET Synth+DA	5.4 ±1.2	10.2 ±2.3	3.9 ±1.1	6.2 ±1.5
Joined lines Donut Synth+HNM+DA	4.3 ±1.0	9.7 ±2.1	4.2 ±1.0	7.9 ±2.0
Separated full-page Donut	8.8 ±1.7	16.3 ±2.9	6.6 ±1.3	12.3 ±2.1
Separated full-page Donut HNM	11.3 ±2.2	20.8 ±3.2	7.7 ±1.6	14.8 ±2.6
Separated full-page Donut DA	7.1 ±1.4	13.1 ±2.7	5.4 ±1.2	10.3 ±2.4
Separated full-page Donut HNM+DA	5.1 ±1.0	9.4 ±2.1	3.9 ±1.0	7.4 ±1.9
Separated full-page Donut Synth+HNM+DA	5.0 ±1.1	9.5 ±2.0	4.0 ±0.9	7.6 ±1.8
Joined full-page Donut	7.1 ±1.7	14.8 ±2.3	6.7 ±1.4	12.8 ±2.0
Joined full-page Donut HNM	9.1 ±1.9	18.9 ±2.6	8.4 ±1.5	16.8 ±2.3
Joined full-page Donut DA	5.7 ±1.4	11.9 ±2.3	5.3 ±1.1	10.1 ±1.9
Joined full-page Donut HNM+DA	4.1 ±1.0	8.6 ±1.9	3.8 ±0.9	7.6 ±2.3
Joined full-page Donut Synth+HNM+DA	4.2 ±1.1	8.7 ±1.9	3.7 ±0.9	7.6 ±2.4

Table 3 Lyrics and music transcription results obtained in the separated lines scenario, the joined lines scenario, the separated full-page scenario and the joined full-page scenario for the Salzinnes dataset

Salzinnes Scenarios	Lyrics Error Rate		Music Error Rate	
	CER	SER	DSER	SER
Separated lines CRNN Synth+DA	3.4 ±0.7	7.4 ±1.2	2.1 ±1.0	5.3 ±1.9
Separated lines CRNN-RNET Synth+DA	3.3 ±0.9	7.4 ±1.2	2.2 ±0.9	5.2 ±1.7
Separated lines Donut Synth+HNM+DA	4.2 ±1.1	9.9 ±2.0	3.7 ±0.9	9.0 ±2.6
Joined lines CRNN Synth+DA	7.0 ±1.7	17.6 ±3.9	2.7 ±1.1	7.0 ±2.1
Joined lines CRNN-RNET Synth+DA	6.7 ±1.6	16.9 ±3.7	2.6 ±1.0	7.2 ±2.2
Joined lines Donut Synth+HNM+DA	3.8 ±0.9	9.1 ±1.8	3.5 ±0.8	8.6 ±2.3
Separated full-page Donut	6.8 ±1.4	12.9 ±2.2	5.6 ±1.3	10.5 ±2.5
Separated full-page Donut HNM	8.1 ±1.6	17.2 ±2.1	7.1 ±1.1	13.2 ±2.2
Separated full-page Donut DA	5.4 ±1.2	10.7 ±2.0	4.9 ±0.9	9.9 ±1.9
Separated full-page Donut HNM+DA	4.0 ±1.0	9.2 ±2.1	3.4 ±0.8	8.3 ±2.5
Separated full-page Donut Synth+HNM+DA	4.1 ±1.0	9.3 ±2.0	3.5 ±0.8	8.4 ±2.5
Joined full-page Donut	6.2 ±1.5	14.2 ±2.1	5.7 ±1.5	13.2 ±2.4
Joined full-page Donut HNM	7.6 ±1.6	16.8 ±2.4	6.9 ±1.2	13.4 ±2.1
Joined full-page Donut DA	5.1 ±1.3	10.2 ±2.1	4.8 ±0.9	9.5 ±1.9
Joined full-page Donut HNM+DA	3.7 ±0.7	8.5 ±1.9	3.2 ±0.8	7.9 ±2.1
Joined full-page Donut Synth+HNM+DA	3.6 ±0.8	8.4 ±1.8	3.2 ±0.9	8.0 ±2.0

- Transformer-based systems obtained very competitive results in the lyrics part (Tables 2 and 3: J. full-page Donut HNM+DA, J. full-page Donut Synth+HNM+DA, J. lines Donut Synth+HNM+DA).
- Transformer-based systems obtained better alignment results than their CRNN-based counterparts. The best results were obtained when the GT with explicit alignment information was provided (Tables 4 and 5: J. full-page Donut HNM+DA).
- Training and decoding at the line level using the Donut model has a slightly lower performance compared to full-page processing. However, the confidence intervals for line-level results largely overlap with those for page-level results, suggesting that this difference is not statistically significant.
- Incorporating synthetic images alongside GT data during Donut model training did not lead to improvements in recognition accuracy. However, this approach helped the model converge faster, enabling the model to achieve

Table 6 Music transcription results obtained in the different scenarios for the Einsiedeln dataset for each of the best performing models comparing geometrical transcription and post processing (GEO) with musical transcription (MUS) and with kern notation transcription (KRN)

Einsiedeln Scenarios	Music DSER	SER
S. lines CRNN Synth+DA GEO	2.9 ±0.8	6.7 ±1.3
S. lines CRNN Synth+DA KRN	3.5 ±1.0	8.4 ±1.6
S. lines CRNN Synth+DA MUS	4.4 ±1.1	10.1 ±1.9
J. lines CRNN Synth+DA GEO	3.9 ±1.0	6.3 ±1.6
J. lines CRNN Synth+DA KRN	4.4 ±1.3	7.8 ±1.8
J. lines CRNN Synth+DA MUS	5.6 ±1.5	9.1 ±2.0
S. full-page Donut HNM+DA GEO	3.9 ±1.0	7.4 ±1.9
S. full-page Donut HNM+DA KRN	4.5 ±1.2	8.9 ±2.2
S. full-page Donut HNM+DA MUS	5.8 ±1.6	10.7 ±2.6
J. full-page Donut HNM+DA GEO	3.8 ±0.9	7.6 ±2.3
J. full-page Donut HNM+DA KRN	4.1 ±1.1	8.5 ±2.4
J. full-page Donut HNM+DA MUS	5.2 ±1.3	9.9 ±2.8

Table 7 Music transcription results obtained in the different scenarios for the and Salzinnes dataset for each of the best performing models comparing geometrical transcription and post processing (GEO) with musical transcription (MUS) and with kern notation transcription (KRN)

Salzinnes Scenarios	Music DSER	SER
S. lines CRNN Synth+DA GEO	2.1 ±1.0	5.3 ±1.9
S. lines CRNN Synth+DA KRN	2.6 ±1.2	6.4 ±2.0
S. lines CRNN Synth+DA MUS	3.1 ±1.4	7.8 ±2.1
J. lines CRNN Synth+DA GEO	2.7 ±1.1	7.0 ±2.1
J. lines CRNN Synth+DA KRN	3.5 ±1.3	8.6 ±2.2
J. lines CRNN Synth+DA MUS	4.1 ±1.4	10.9 ±2.6
S. full-page Donut HNM+DA GEO	3.4 ±0.8	8.3 ±2.5
S. full-page Donut HNM+DA KRN	4.1 ±1.0	10.2 ±2.5
S. full-page Donut HNM+DA MUS	5.2 ±1.1	12.1 ±2.9
J. full-page Donut HNM+DA GEO	3.2 ±0.8	7.9 ±2.1
J. full-page Donut HNM+DA KRN	4.3 ±1.2	9.9 ±2.7
J. full-page Donut HNM+DA MUS	5.3 ±1.3	11.8 ±2.8

Table 8 Music and lyrics transcription results obtained for each of the best performing experimental setups in the separated (S.) full-page scenario and the joined (J.) full-page scenario for the Einsiedeln dataset with narrative texts

Einsiedeln Narrative Scenarios	Music DSER	Lyrics CER
S. full-page Donut HNM+DA	4.0 ±1.1	6.2 ±1.3
S. full-page Donut Synth+HNM+DA	3.9 ±1.0	6.1 ±1.2
J. full-page Donut HNM+DA	3.8 ±0.9	5.9 ±1.1
J. full-page Donut Synth+HNM+DA	3.8 ±1.0	6.0 ±1.1

Table 9 Music and lyrics transcription results obtained for each of the best performing experimental setups in the separated (S.) full-page scenario and the joined (J.) full-page scenario for the Salzinnes dataset with narrative texts

Salzinnes Narrative Scenarios	Music DSER	Lyrics CER
S. full-page Donut HNM+DA	4.1 ±1.0	6.0 ±1.1
S. full-page Donut Synth+HNM+DA	4.0 ±1.1	6.1 ±1.3
J. full-page Donut HNM+DA	3.7 ±0.8	5.8 ±0.9
J. full-page Donut Synth+HNM+DA	3.8 ±0.8	5.7 ±0.9

comparable transcription quality with reduced computational time.

- The limited amount of data did not hinder the model's ability to learn how to produce musical and lyrical syllables. This was despite the model having to learn how to structure the transcriptions from English words to musical and lyrical syllables.
- Regardless of the size, the transformer-based system outperformed the CRNN-based system regarding the AMNLT task, taking into account both transcription quality and alignment quality.
- When narrative texts were present as noise in full-page images, performance was worse with respect to configurations where such texts were removed. This effect is attributed to the model's tendency to partially recognize these elements despite their exclusion from the GT.
- The use of a ResNet backbone did not result in improvements in transcription quality compared to the simpler CNN backbone. This suggests that the basic model is capable of fully capturing the intricacies of the datasets and the more relevant aspect lies in the relationships between these elements.

It can also be observed that there is a considerable overlap in the confidence intervals in both the lyrics and music recognition results. This means that while the improvements in alignment are significant, the differences in the results obtained in transcription are not critical according to the confidence intervals.

In regards to the comparison of results when using the Donut model at the line and at the full-page levels, due to the Donut model's robust architecture as well as the use of fine-tuning techniques and data augmentation, the model has been able to adapt to the different structures presented to it. The slight drop in performance at the line level can be attributed to the reduced contextual information available, which limits the model's ability to learn compared to when it is trained and evaluated at the page level. This outcome was consistent with expectations.

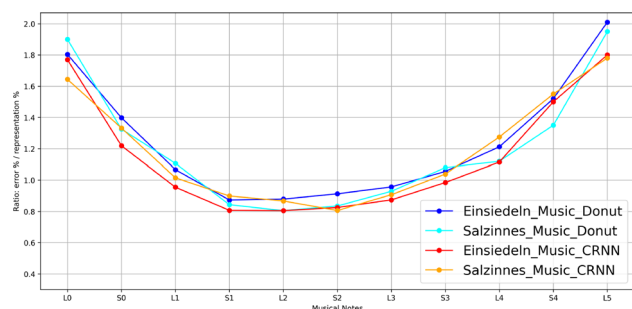


Fig. 7 Music ratio of errors percentage to representation percentage in the GT

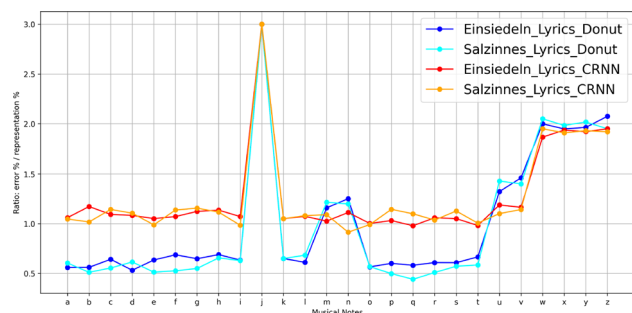


Fig. 8 Lyrics ratio of errors percentage to representation percentage in the GT

With regard to the post-processing of the geometrical positions to obtain musical notes, 98% of the musical clefs were recognized correctly. Given that the translation from geometrical to musical notation will only introduce mistakes when the clefs are not properly recognized. This process was successful in almost every single instance in our experiments. If the clefs were recognized with 100% accuracy, the results regarding alignment and musical transcription would be about 4% better, a minor difference in results, as there would be no mistakes derived from the post-processing step.

A more detailed analysis of the types of errors produced by our model reveals a similar number of insertions and deletions across the different data partitions. Furthermore, the number of substitutions is approximately equal to the combined total of insertions and deletions when computing CER, DSER, and SER. An analysis of character-level relative error provides additional insights. Figures 7 and 8 depict the ratio of each character's presence in the errors relative to its representation in the GT. A ratio greater than one indicates that a character contributes disproportionately to errors compared to its frequency in the dataset.

As shown in Fig. 7, musical symbols exhibit similar error distributions for both the CRNN and Donut models. The most error-prone elements are those located at the extremes of the staff, which correspond to the least frequent symbols in the dataset. This observation suggests that limited representation during training negatively impacts recognition accuracy.



Fig. 9 Examples of the words "martirij" and "ieiunjis" extracted from the document, where in both cases the letter "j" (highlighted in red) was misrecognized as the letter "i". As a result, the words were transcribed as "martirii" and "ieiuniis", respectively



Fig. 10 Examples of the words "nosmet" and "nunc" extracted from the document, which were misrecognized as "nosnet" and "mun", respectively. In both cases, a character (highlighted in red) was mistaken for the opposite one ("m" for "n" and viceversa)

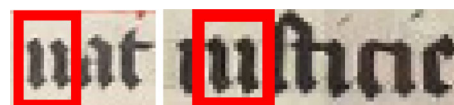


Fig. 11 Examples of the words "vat" and "iusticie" extracted from the document, which were misrecognized as "uat" and "ivsticie", respectively. In both cases, a character (highlighted in red) was mistaken for the opposite one ("u" for "v" and viceversa)

Clef data is not included as its uniform distribution gave no additional insight to errors. The similarity in error patterns between CRNN and Donut aligns with their comparable transcription performance.

Conversely, Fig. 8 reveals distinct error distributions for lyrical characters across the two models. For both models, letters with low frequency in the dataset (e.g., "w", "x", "y", and "z") exhibit high error ratios. Additionally, the letter "j" shows a disproportionately high error rate, likely due to its visual similarity to "i", which is far more common; in many cases, "j" is misclassified as "i". Data regarding capital letters is not included as their representation gave no additional insight. Beyond these cases, CRNN errors are relatively evenly distributed across the alphabet, indicating a similar difficulty in recognizing most characters.

In contrast, the Donut model exhibits concentrated errors in a few specific letters, such as "m" and "n", and "u" and "v", which are often confused due to their visual similarity. The fact that Donut exhibits concentrated errors in a few problematic letters, whereas CRNN shows errors distributed across most of the vocabulary could be the main indicator of the Donut model having a better performance regarding lyric recognition. Donut's difficulties are limited to characters with low representation or inherent ambiguity, while CRNN struggles with character recognition are more broad.

Figures 9, 10, and 11 illustrate examples of characters where recognition errors are concentrated for both the CRNN model and, more notably, the Donut model in instances where they were not properly recognized as their own character.

In addition, syllabic separators are recognized with accuracy. These separators constitute approximately 20% of the symbols in these datasets and contribute to only 10% of the total errors. This suggests that errors related to segmentation are not primarily due to difficulties in recognizing these symbols.

Overall, this would indicate that while there is not a clear origin for the different mistakes caused by the model, there is not a systematic flaw in the model either. This supports the conclusion that the model is functioning as intended. To further expand on this point, the experiments we have conducted with the CRNN model at the line level and with the Donut model at both line and full-page levels under equivalent conditions (which included the use of synthetic images, testing with both a simple CNN backbone and a more robust pre-trained ResNet backbone to approximate the architecture of the Donut model, and applying identical experimental setups for CRNN and Donut at the line level) and based on these comparisons, we can conclude that the observed differences in performance primarily stem from the inherent architectural characteristics of each model, their learning processes, and their unique capabilities.

With regard to the alignment error, it can be observed that there is a notable improvement over the baseline error with respect to both the naive and the IBM Model 2 alignment approaches. About a 50% improvement over the best baseline results was obtained for both CRNN-based and transformer-based systems in the Einsiedeln dataset. A 65% improvement for CRNN-based systems and a 77% improvement for transformer-based systems over the best baseline results were obtained in the Salzinnes dataset.

This shows the ability of these CRNN and Transformer models for learning the alignments if properly instructed. For the case of transformer-based models, they not only obtained the best alignment results, but they also maintained good transcription quality. All of this shows that these transformer-based systems offer a great solution for the AMNLT challenge and for the OMR task as a whole.

5 Conclusions

In this work, we have proposed two methods to approach the AMNLT task for OMR. The systems were able to produce both a high quality transcription of a musical document and the alignment between the musical and lyrical elements. We explored how these systems performed on two datasets from a similar period. We evaluated the transcription and alignment quality for these datasets.

Great improvements in alignment quality were found when considering training with aligned samples; from 50% to 77% in alignment error in both the Einsiedeln and Salzinnes datasets. These findings emphasize the critical role of align-

ment training in improving alignment quality, making it a valuable technique when working with these historical musical documents. We have also observed that when alignments are considered, the difference in results for transcription quality are not significant. In the case of lyrics recognition, transformer-based systems, outperformed CRNN-based systems and, in the case of music recognition, there were mixed results, favouring CRNN-based systems.

One interesting result is that when datasets are smaller, transformer-based systems outperform CRNN in every metric for the AMNLT task, contrary to what is usual for models of this kind in other tasks. This is because transformer-based systems are better suited than CRNN for the AMNLT task, given that the CRNN are not prepared for the transcription of simultaneous symbols given a singular time frame. It is only when datasets are bigger that results with CRNN-based systems are able to learn the intricacies of alignments and obtain similar results to transformer-based systems.

This is one of the reasons that transformer-based systems are ideal for this task, together with the fact that we can avoid the segmentation and extraction of lines and that no pre-processing need be applied to them. They are a very suitable model for the AMNLT task.

Future research could include being able to produce more data (mainly the generation of more adequate synthetic data) that fits the training distribution to improve upon results. Another important point that we could derive from this work is to try to utilize the output of transformer-based systems to perform not only transcription recognition but to also perform information search over the transcriptions obtained. With this, we could perform the indexation of elements, which would allow us to develop search systems and databases that currently can only be developed using classical methods.

As final remarks, the results show that the techniques presented in this work provide a high level of quality alignment while maintaining the quality of transcriptions, allowing for a segmentation-free approach to solve the AMNLT problem. This is a great step for the OMR community.

Appendix: complementary results

This appendix includes complementary results on a different dataset to support some of the claims made throughout the article. This dataset is the VORAU dataset, which does not present alignments in its GT and therefore it is not used in the AMNLT experiments. However, it remains a valuable point of comparison for evaluating the recognition capabilities of the models presented in this study. The characteristics of the VORAU dataset can be seen in Table 10. (See Fig. 12)

This dataset has been split into five partitions for cross validation, where 80% of the data was used for training and validation and the remaining 20% was used for testing. For

Table 10 Characteristics of the VORAU dataset

Dataset	VORAU
Number of pages	492
Number of pairs staff/lyrics	4541
Number of narrative lines	520
Number of music symbols	29
Number of lyrics symbols	51

**Fig. 12** Example pages from the VORAU dataset**Table 11** Lyrics transcription results obtained in the separated (S.) lines scenario and the separated full-page scenario for the VORAU dataset

VORAU Lyrics Scenarios	Lyrics CER	WER
S. lines CRNN Synth+DA	9.4 ±2.3	36.6 ±7.2
S. lines CRNN-RNET Synth+DA	9.3 ±2.4	36.1 ±7.4
S. lines Donut Synth+HNM+DA	7.9 ±2.3	33.2 ±9.2
S. full-page Donut	10.1 ±2.5	37.4 ±10.6
S. full-page Donut HNM	13.1 ±3.1	40.2 ±11.1
S. full-page Donut DA	9.2 ±2.4	35.1 ±9.9
S. full-page Donut HNM+DA	7.5 ±2.2	32.7 ±9.0
S. full-page Donut Synth+HNM+DA	7.7 ±2.1	33.1 ±9.3

the music part DSER will be evaluated, for the lyrics part CER will be evaluated. Additionally to CER for lyrics, Word Error Rate (WER) will be used for this dataset, as instead of the lyrical elements being segmented in syllables, they are segmented at the word level. SER was not evaluated for either dataset, as syllabic segmentation is not provided. As with the other two datasets, the evaluation was performed across all five partitions, and a 95% confidence interval was computed using bootstrapping.

Tables 11 and 12 show the transcription results obtained in the experiments for the separated scenarios, as these are the only applicable scenarios for this dataset.

Table 12 Music transcription results obtained in the separated (S.) lines scenario and the separated full-page scenario for the VORAU dataset

VORAU Music Scenarios	Music DSER
S. lines CRNN Synth+DA	1.9 ±0.9
S. lines CRNN-RNET Synth+DA	2.0 ±0.9
S. lines Donut Synth+HNM+DA	3.2 ±1.0
S. full-page Donut	3.9 ±1.1
S. full-page Donut HNM	4.5 ±1.2
S. full-page Donut DA	3.3 ±1.0
S. full-page Donut HNM+DA	2.9 ±0.8
S. full-page Donut Synth+HNM+DA	2.8 ±0.9

These results present an even more pronounced version of some of the patterns observed in the experiments conducted on the Einsiedeln and Salzinnes datasets. It is evident that the experimental setup aligns with the trends previously identified. Furthermore, the base models demonstrate sufficient capacity to extract the most effective representation of the data. In this case, transcription quality is clearly differentiated: the Donut model outperforms the CRNN in lyrics recognition, while the CRNN model shows a better performance in music recognition. However, it is worth noting that the confidence intervals for both cases overlap.

Although these results do not directly expand on the main focus of this article, the task of AMNLT, they help reinforce the findings regarding the distinct behaviors of the CRNN and Donut models, and how their performance is tied to the differences in their architectures, learning processes, and the type of structures each model is best suited to handle.

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Data Availability Data references are provided within the manuscript.

Declarations

Competing interests The authors declare no competing interests.

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